

Chapter 4:

Networking and the Internet

Computer Science: An Overview
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Chapter 4: Networking and the Internet

- 4.1 Network Fundamentals
- 4.2 The Internet
- 4.3 The World Wide Web
- 4.4 Internet Protocols
- 4.5 Security

Network Classifications

- Scope
 - Personal area network (PAN)
 - Local area network (LAN)
 - Metropolitan area (MAN)
 - Wide area network (WAN)
- Ownership
 - Closed versus open
- Topology (configuration)
 - Bus (Ethernet)
 - Star (Wireless networks with central Access Point)

Figure 4.1 Network topologies

a. Bus

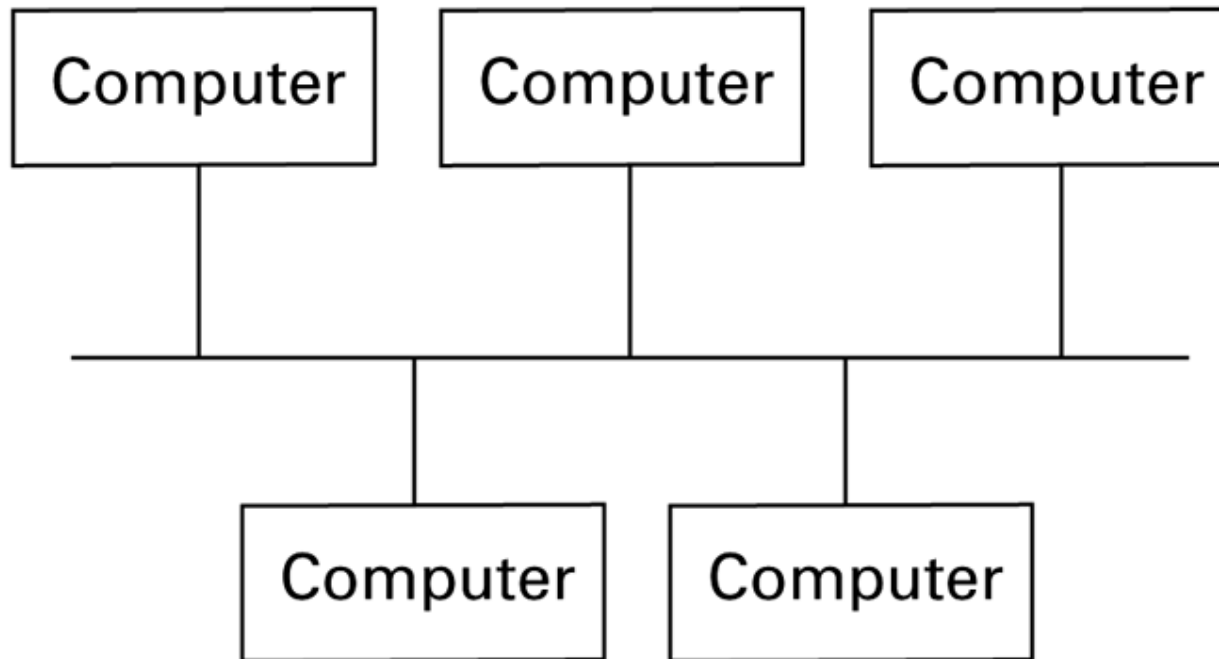
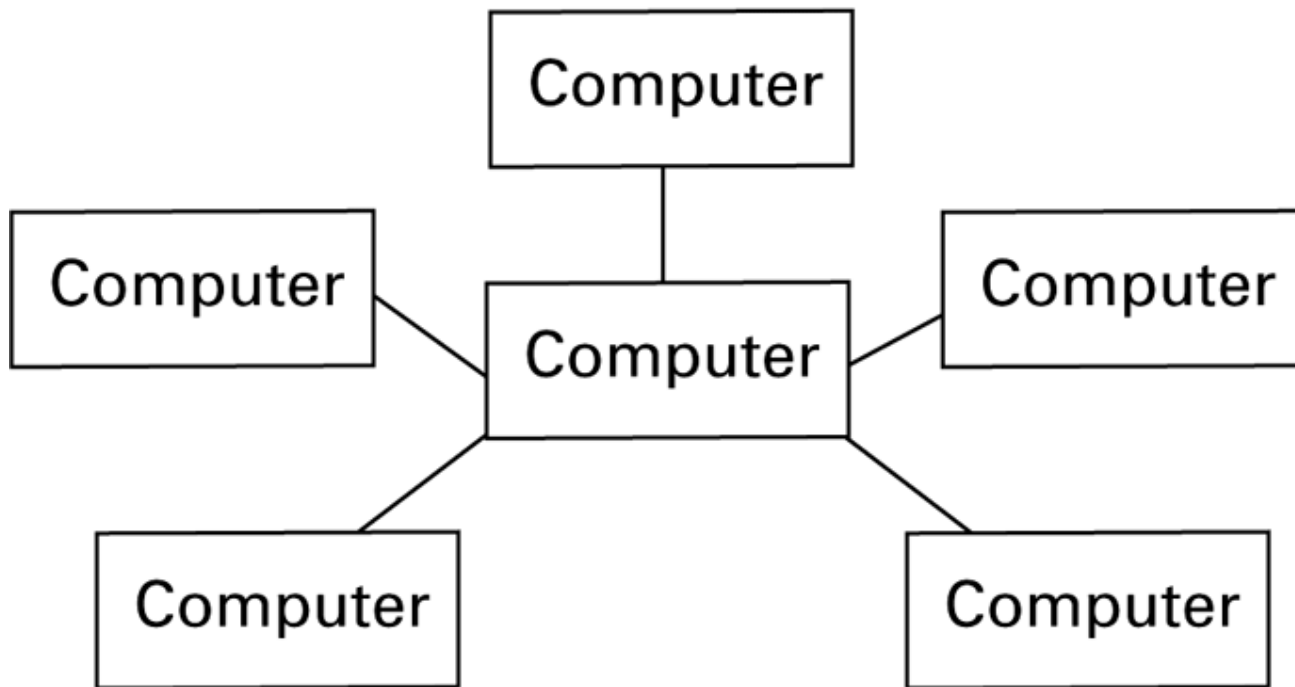


Figure 4.1 Network topologies (continued)

b. Star



Protocols

- Carrier Sense Multiple Access/Collision Detection (CSMA/CD)
 - Used in Ethernet
 - Silent bus provides right to introduce new message
- Carrier Sense Multiple Access/Collision Avoidance (CSMA/CA)
 - Used in WiFi
 - Hidden terminal problem

Figure 4.2 Communication over a bus network

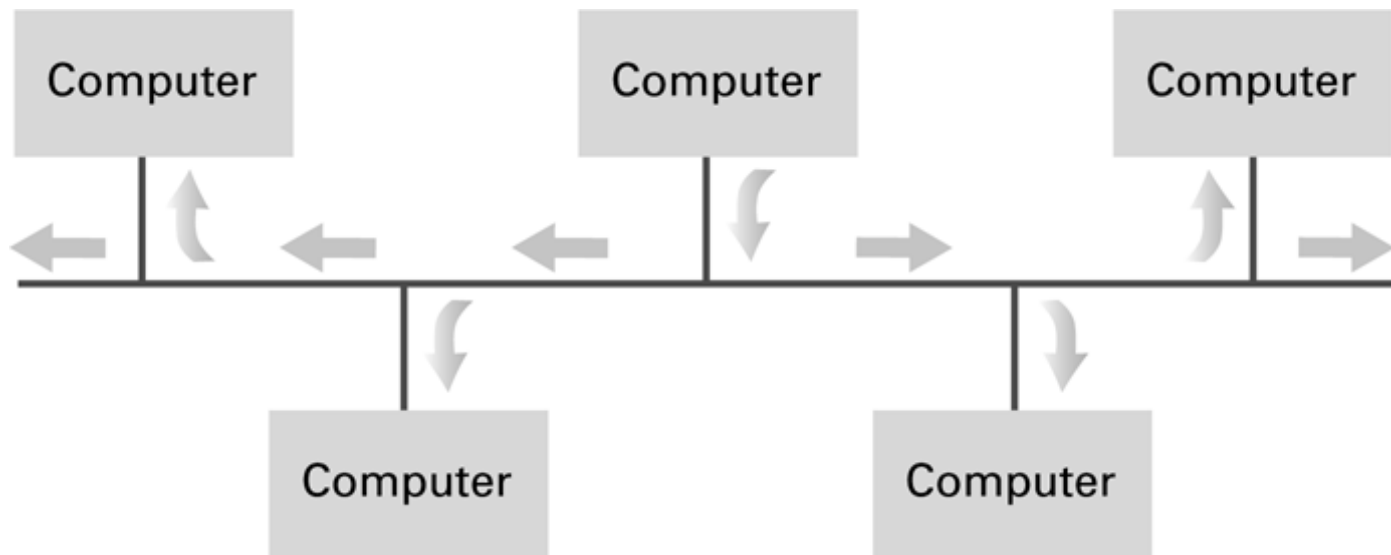
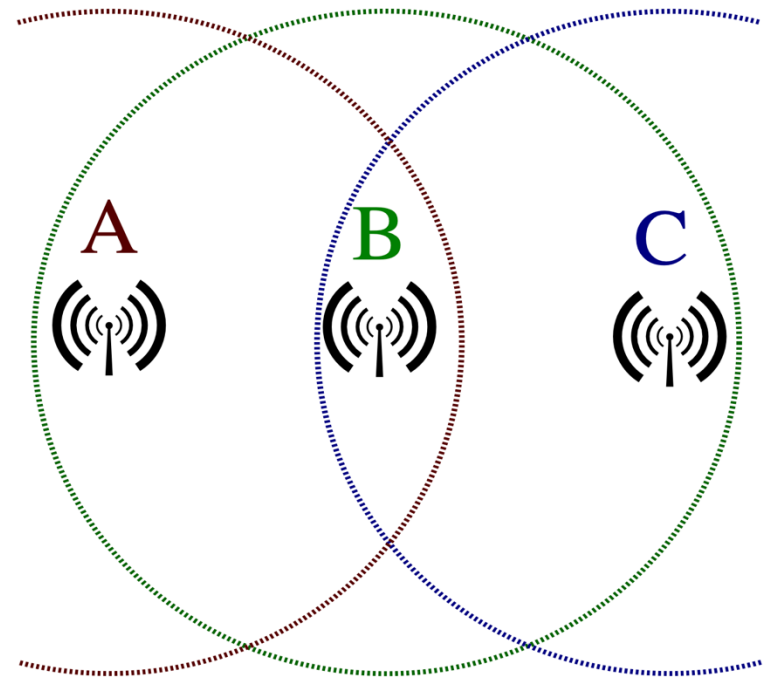


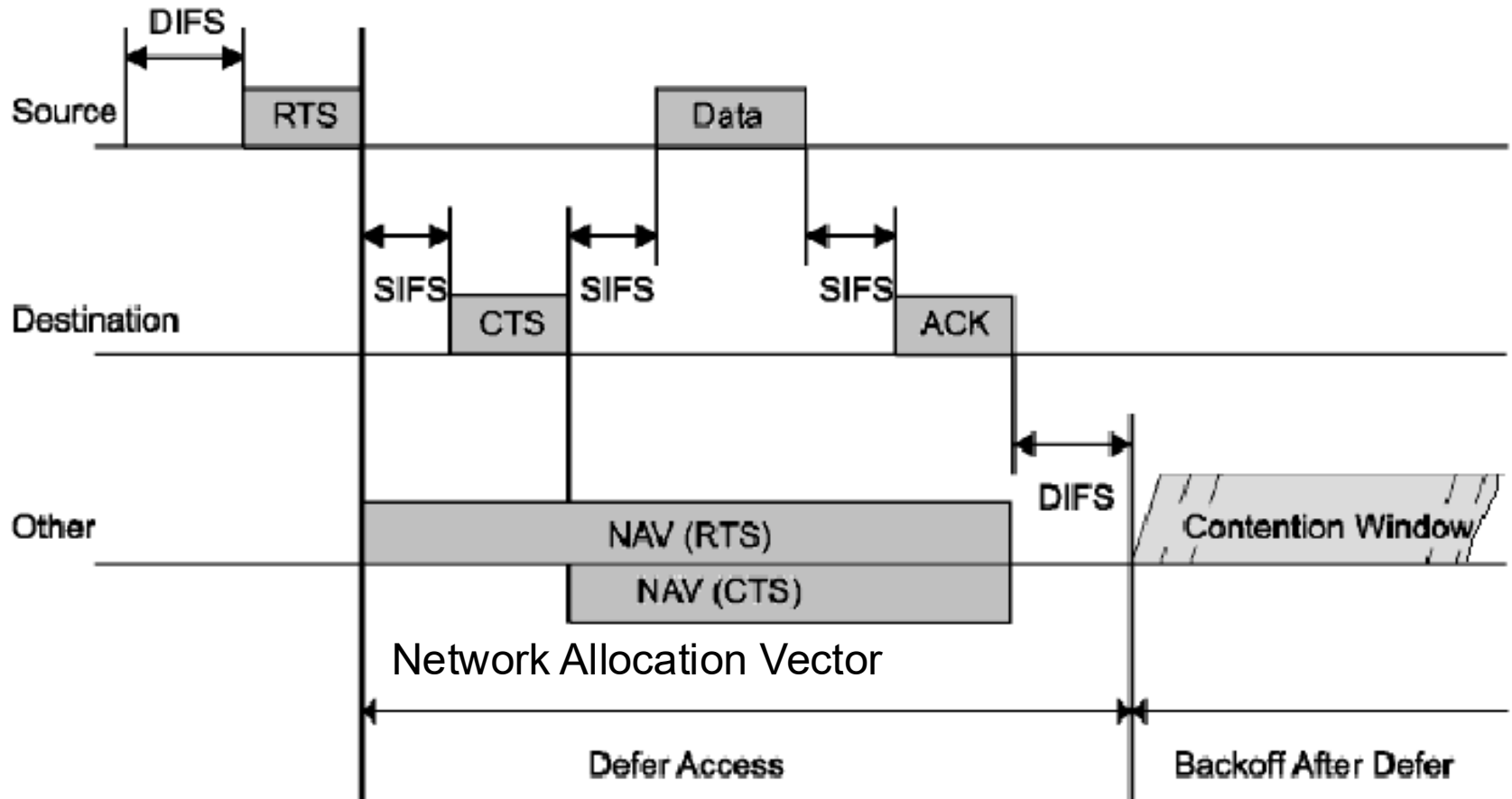
Figure 4.3 The hidden terminal problem

Scenario

1. Imagine three nodes: **Node A**, **Node C**, and a common receiving node **Node B**.
2. **Node A** can communicate with **B**, and **Node C** can communicate with **B**.
3. However, **Node A** and **Node C** are far apart or separated by an obstacle (like a wall), so they are out of each other's radio range and cannot hear each other's transmissions. **A** is "hidden" from **C**, and **C** is "hidden" from **A**.
4. **Node A** wants to send data to **B**. It senses the channel, finds it idle (because it cannot hear **C**), and begins transmitting.
5. Simultaneously, **Node C** also wants to send data to **B**. It senses the channel, finds it idle (because it cannot hear **A**), and begins transmitting.
6. The signals from both **A** and **C** arrive at **B** at the same time, resulting in a collision. The data received by **B** is corrupted and must be discarded, forcing **A** and **C** to retransmit later.



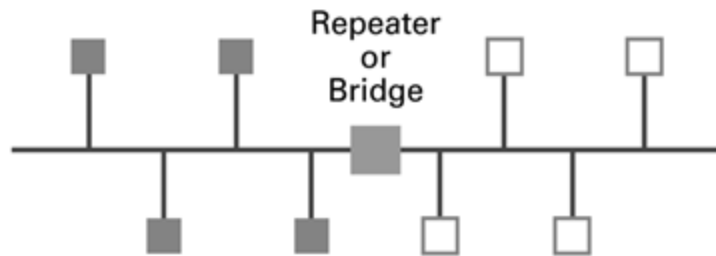
Request-to-send/Clear-to-send (RTS/CTS) handshaking



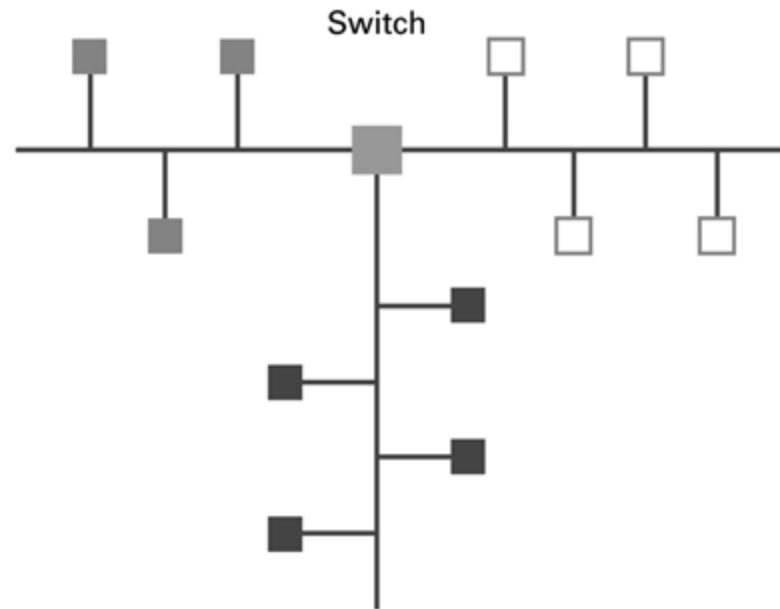
Connecting Networks

- **Repeater:** Extends a network
- **Bridge:** Connects two compatible networks
- **Switch:** Connects several compatible networks
- **Router:** Connects two incompatible networks resulting in a network of networks called an **internet**

Figure 4.4 Building a large bus network from smaller ones

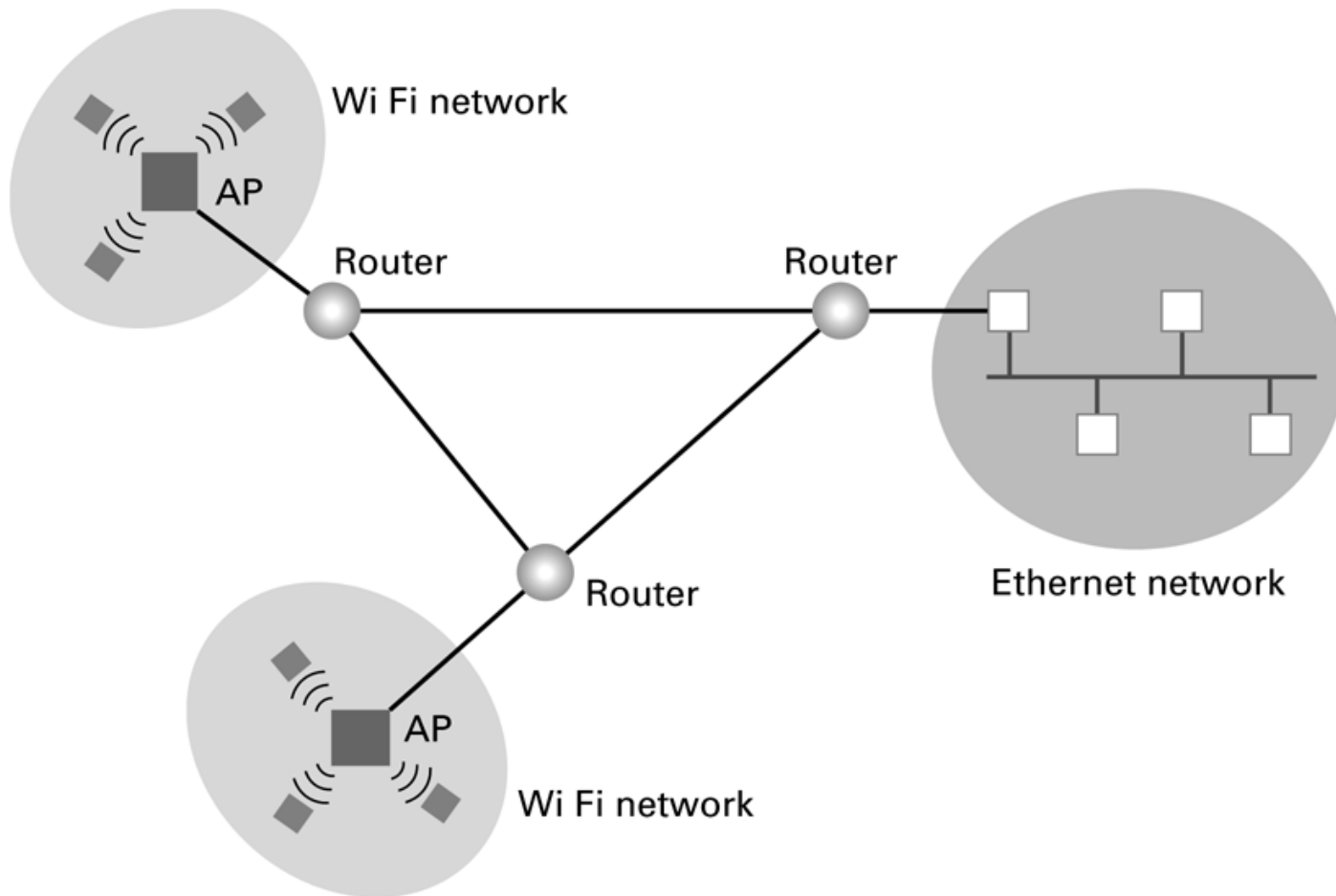


a. A repeater or bridge connecting two buses



b. A switch connecting multiple buses

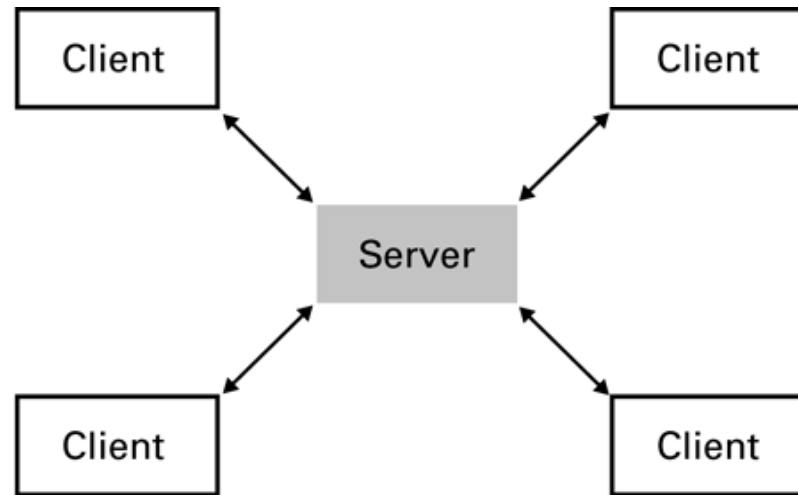
Figure 4.5 Routers connecting two WiFi networks and an Ethernet network to form an internet



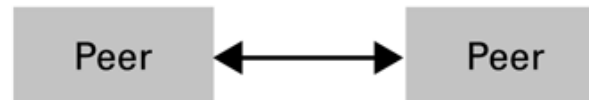
Inter-process Communication

- Client-server
 - One server, many clients
 - Server must execute continuously
 - Client initiates communication
- Peer-to-peer (P2P)
 - Two processes communicating as equals
 - Peer processes can be short-lived

Figure 4.6 The client/server model compared to the peer-to-peer model



a. Server must be prepared to serve multiple clients at any time.



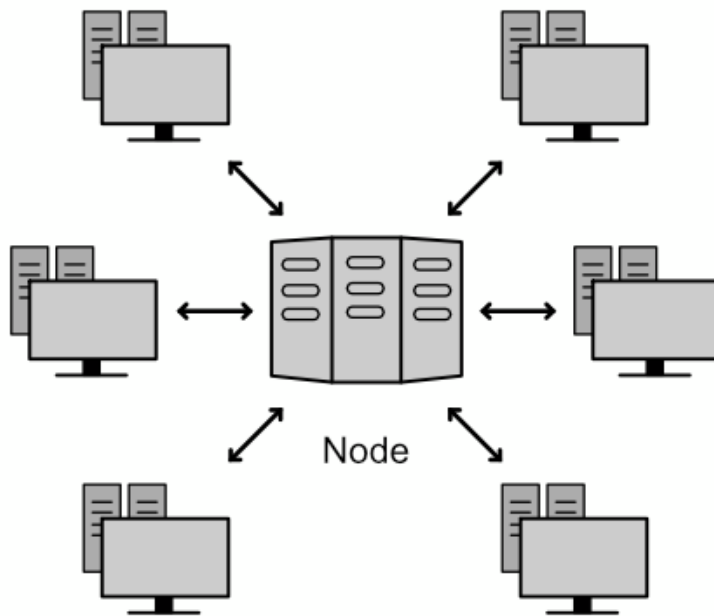
b. Peers communicate as equals on a one-to-one basis.

Distributed Systems

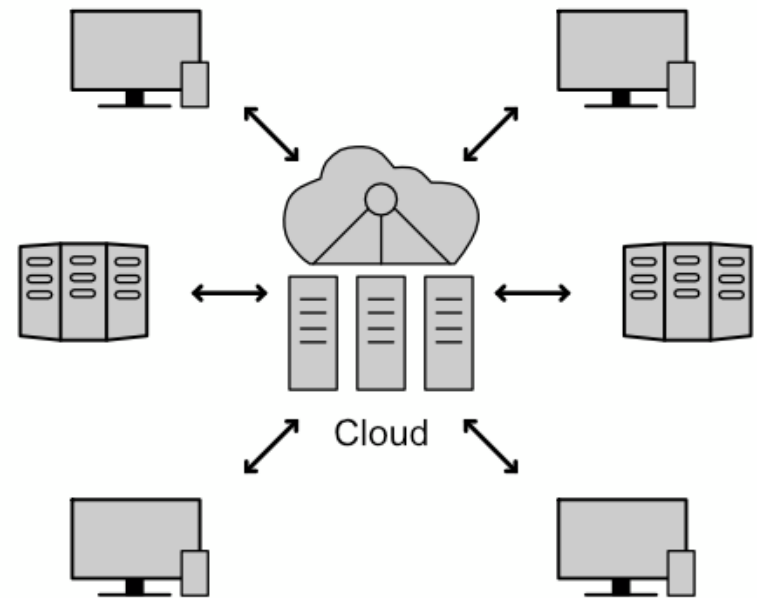
- Systems with parts that run on different computers
 - Cluster computing
 - Grid computing
 - Cloud computing
 - Amazon's Elastic Compute Cloud
 - Google Drive

Distributed Systems

Grid Computing



Cloud Computing



The Intranet

- A private, secure network for a single organization to share information, improve communication, and facilitate collaboration among employees.

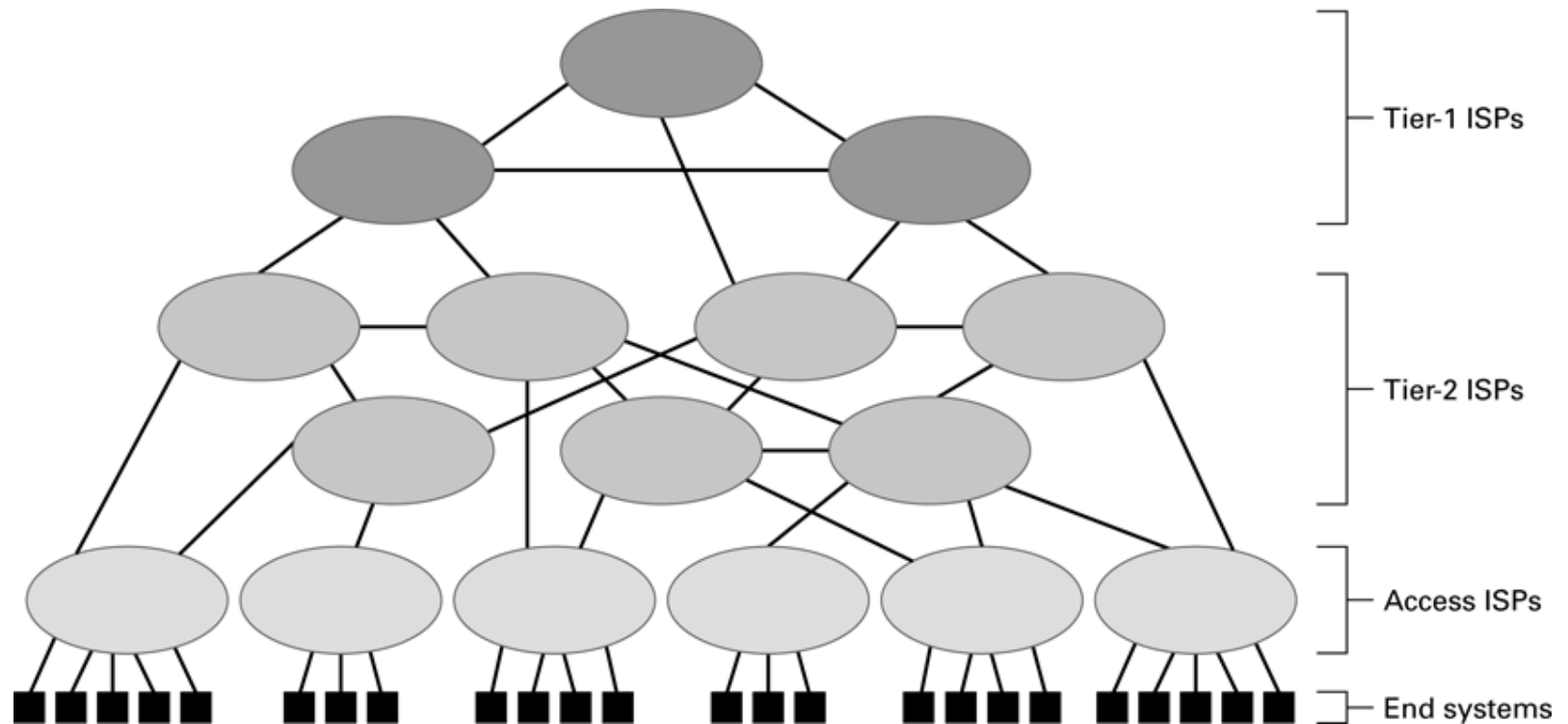
The Internet

- The Internet: An internet that spans the world
 - Original goal was to develop a means of connecting networks that would not be disrupted by local disasters
 - Today a commercial undertaking that links a worldwide combination of PANs, LANs, MANs, and WANs involving millions of computers

Internet Architecture

- Internet Service Provider (ISP)
 - Tier-1
 - Tier-2
- Access or tier-3 ISP: Provides connectivity to the Internet
 - Hot spot (wireless)
 - Telephone lines
 - Cable/Satellite systems DSL
 - Fiber optics

Figure 4.7 Internet Composition



IPv5 Address Classes and Ranges

IPv4 Address Classes and Ranges						
Address Class	Type	Range	Default Subnet Mask	Number of Networks	No of Hosts Per Network	Use
A	Public	1.0.0.0 to 127.0.0.0	255.0.0.0	126	16,777,214	Governments and Large Number of Hosts
	Private	10.0.0.0 to 10.255.255.255				
B	Public	128.0.0.0 to 191.255.255.255	255.255.0.0	16,382	65,534	Medium Companies
	Private	172.16.0.0 to 172.31.255.255				
C	Public	192.0.0.0 to 223.255.255.255	255.255.255.0	2,097,150	254	Small Companies and LANs
	Private	192.168.0.0 to 192.168.255.255				
D	N/A	224.0.0.0 to 239.255.255.255	Not Applicable	N/A	N/A	Reserved for Multicasting
E	N/A	240.0.0.0 to 254.255.255.255	Not Applicable	N/A	N/A	Experimental
Special	Special	127.0.0.1 to 127.255.255.255	N/A	N/A	N/A	Loopback Testing
Note: - Addresses 127.0.0.1 to 127.255.255.255 cannot be used and are reserved for loopback testing - APIPA address range is 169.254.0.1 to 169.254.255.254 and has 65, 534 usable IP addresses, with the subnet mask of 255.255.0.0.						

Internet Addressing

- **IP address:** pattern of 32 or 128 bits often represented in dotted decimal notation
- **MAC address:** 12-character alphanumeric code that is permanently assigned by the manufacturer and used to identify devices on a local network. **The first six characters** of the MAC address identify the manufacturer (Organizationally Unique Identifier or OUI).
- Mnemonic address:
 - Domain names
 - Top-Level Domains
- Domain name system (DNS)
 - Name servers
 - DNS lookup

Internet Corporation for Assigned Names & Numbers (ICANN)

- Allocates IP addresses to ISPs who then assign those addresses within their regions.
- Oversees the registration of domains and domain names.

Early Internet Applications

- Network News Transfer Protocol (NNTP)
- File Transfer Protocol FTP (port 21)
- Telnet and SSH (port 22)
- Hypertext Transfer Protocol (HTTP) (port 80)
- Hypertext Transfer Protocol Secure (HTTPS) (port 443)
- Internet Control Message Protocol (ICMP) used to send error messages and operational information
- Electronic Mail (email)
 - Domain mail server collects incoming mail and transmits outgoing mail
 - Mail server delivers collected incoming mail to clients via POP3 (Post Office Protocol version 3) or IMAP (Internet Message Access Protocol)

SMTP Simple Mail Transfer Protocol

220 mail.tardis.edu SMTP Sendmail Gallifrey-1.0; Fri, 23 Aug 2413 14:34:10

HELO mail.skaro.gov

250 mail.tardis.edu Hello mail.skaro.gov, pleased to meet you

MAIL From: dalek@skaro.gov

250 2.1.0 dalek@skaro.gov... Sender ok

RCPT To: doctor@tardis.edu

250 2.1.5 doctor@tardis.edu... Recipient ok

DATA

354 Enter mail, end with "." on a line by itself

Subject: Extermination.

EXTERMINATE!

Regards, Dalek

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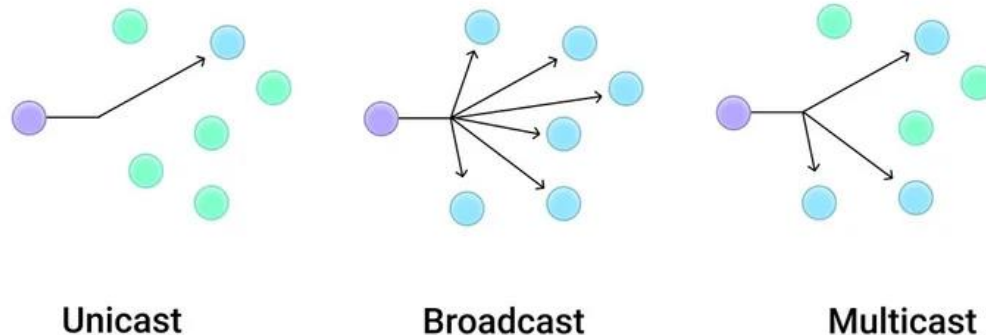
250 2.0.0 r7NJYAE1028071 Message accepted for delivery

QUIT

221 2.0.0 mail.tardis.edu closing connection

More Recent Applications

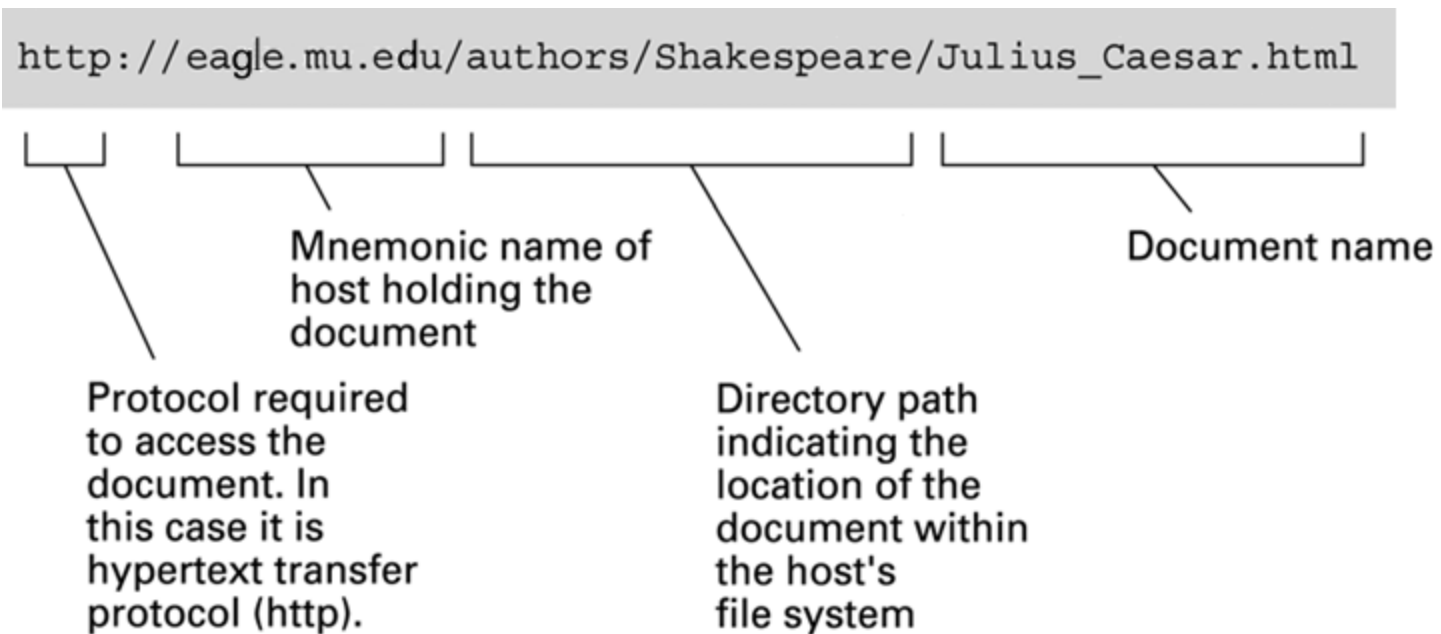
- Voice Over IP (VoIP)
- Internet Multimedia Streaming
 - N-unicast
 - Multicast
 - On-demand streaming
 - Content delivery networks (CDNs)



World Wide Web

- **Hypertext** combines internet technology with concept of linked-documents
 - Embeds **hyperlinks** to other documents
- **Browsers** present materials to the user
- **Webservers** provide access to documents
- Documents are identified by **URLs** and transferred using **HTTP**

Figure 4.8 A typical URL



Hypertext Markup Language (HTML)

- Encoded as text file
- Contains tags to communicate with browser
 - Appearance
 - `<h1>` to start a level one heading
 - `<p>` to start a new paragraph
 - Links to other documents and content
 - `<a href = . . . >`
 - Insert images
 - `<img src = . . . >`

Figure 4.9 A simple webpage

a. The page encoded using HTML.

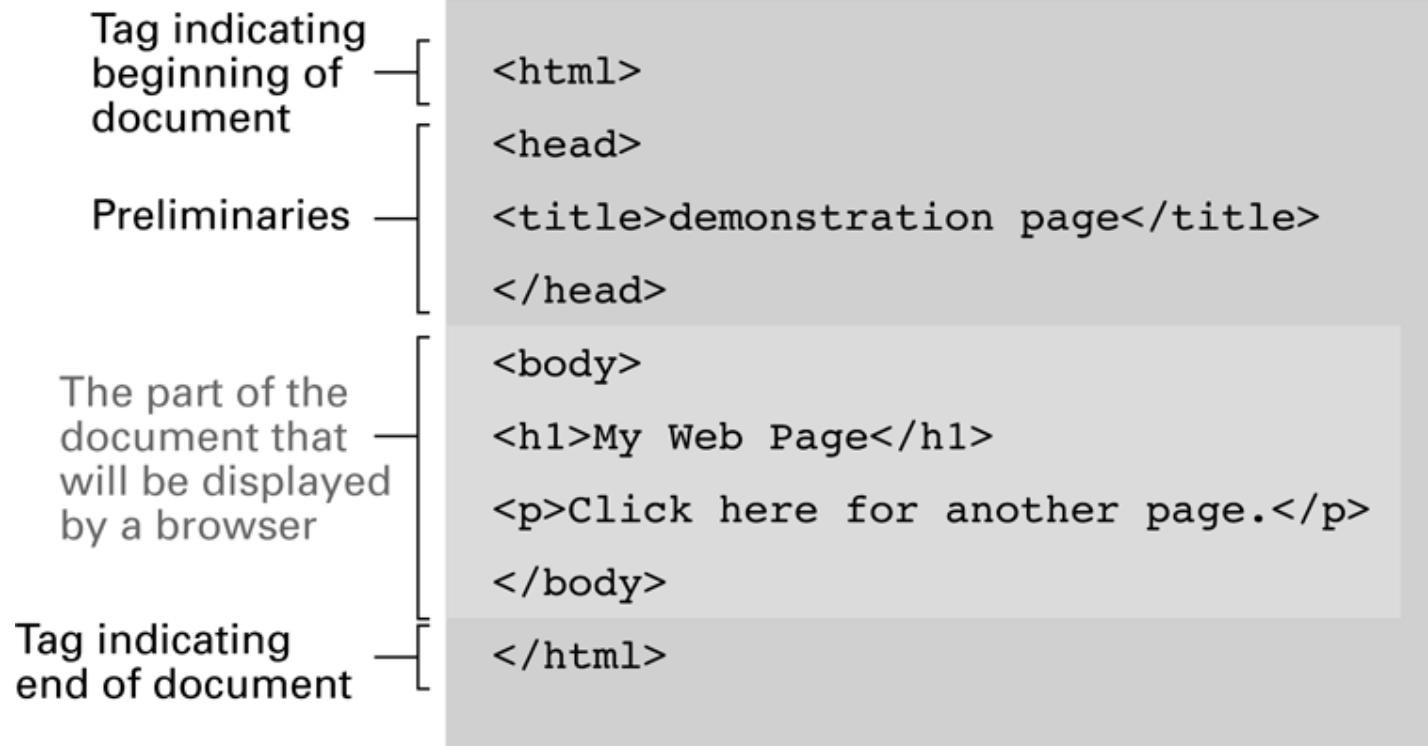


Figure 4.9 A simple webpage (continued)

b. The page as it would appear on a computer screen.



Figure 4.10 An enhanced simple webpage

a. The page encoded using HTML.

Anchor tag
containing
parameter — [

Closing
anchor tag — [

```
<html>
<head>
<title>demonstration page</title>
</head>
<body>
<h1>My Web Page</h1>
<p>Click
  <a href="http://crafty.com/demo.html">
    here
  </a>
  for another page.</p>
</body>
</html>
```


Figure 4.10 An enhanced simple Web page (continued)

b. The page as it would appear on a computer screen.



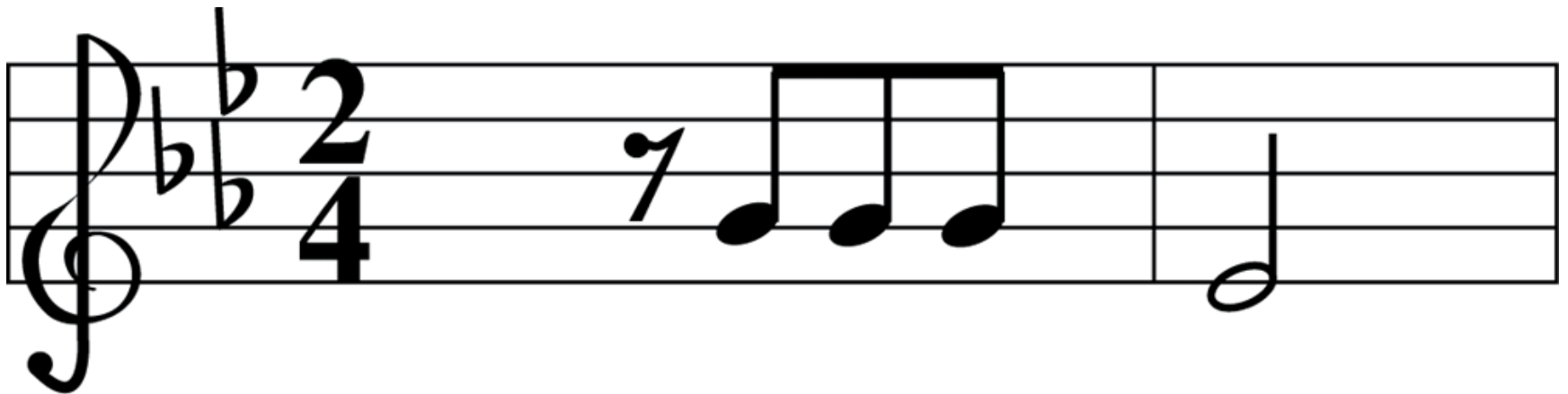
Extensible Markup Language (XML)

- XML: A language for constructing markup languages similar to HTML
 - A descendant of SGML
 - Opens door to a World Wide *Semantic* Web

Using XML

```
<staff clef = "treble"> <key>C minor</key>  
<time> 2/4 </time>  
<measure> < rest> egth </rest> <notes>  
    egth G, egth G, egth G  
    </notes></measure>  
<measure> <notes> hlf E  
    </notes></measure>  
</staff>
```

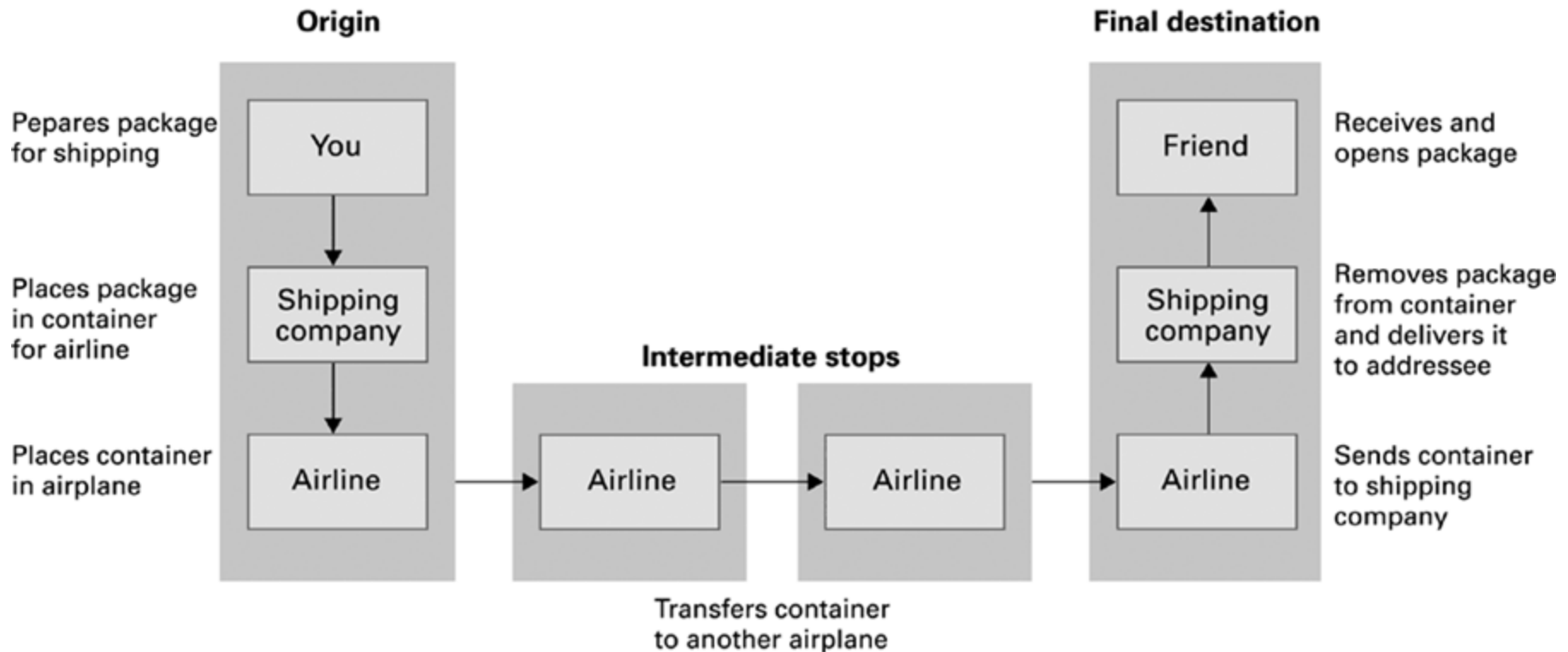
Figure 4.11 The first two bars of Beethoven's Fifth Symphony



Client Side Versus Server Side

- Client-side activities
 - Javascript
 - Macromedia Flash
- Server-side activities
 - Common Gateway Interface (CGI)
 - Servlets
 - JavaServer Pages (JSP) / Active Server Pages (ASP)
 - PHP

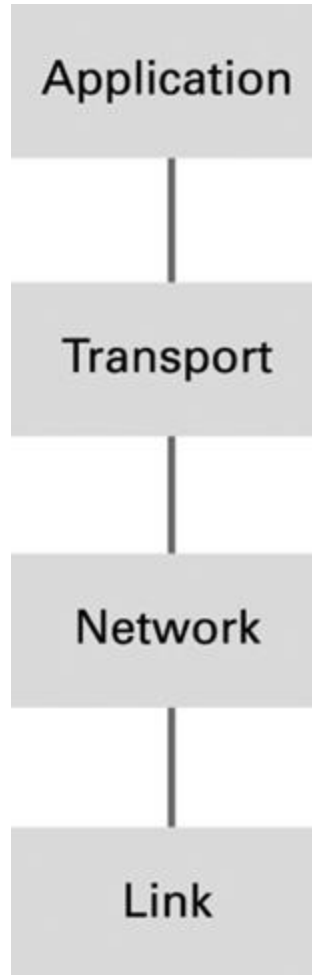
Figure 4.12 Package-shipping example



Internet Software Layers

- **Application:** Constructs message with address
- **Transport:** Chops message into packets
- **Network:** Handles routing through the Internet
- **Link:** Handles actual transmission of packets

Figure 4.13 The Internet software layers



The OSI (Open Systems Interconnection) model

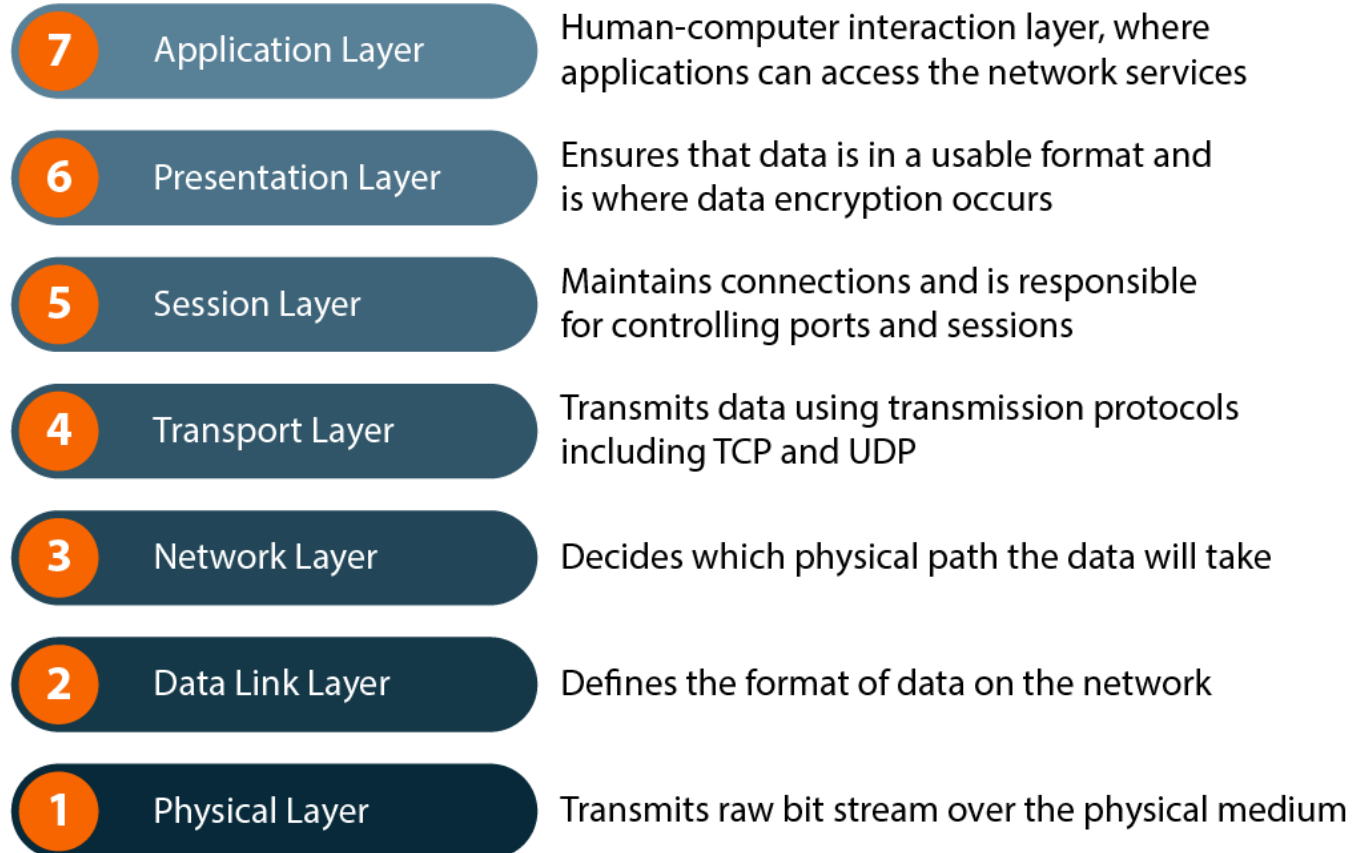
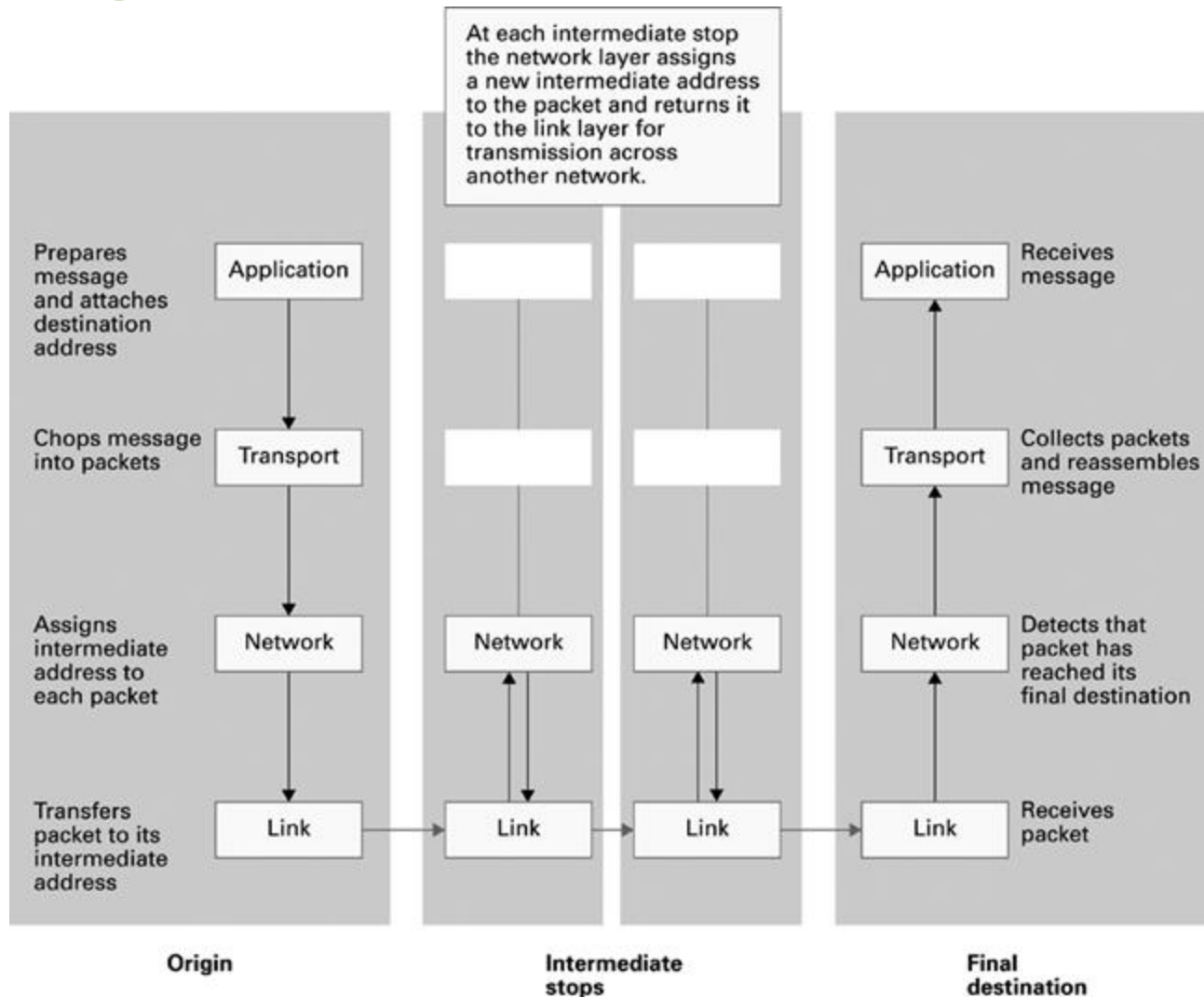


Figure 4.14 Following a message through the Internet

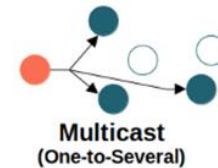
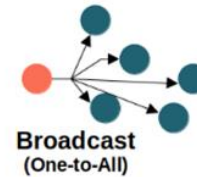
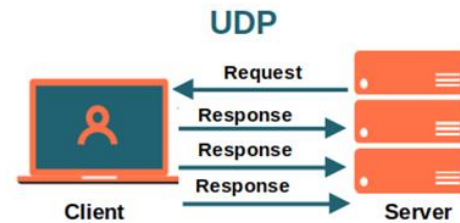
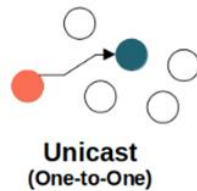
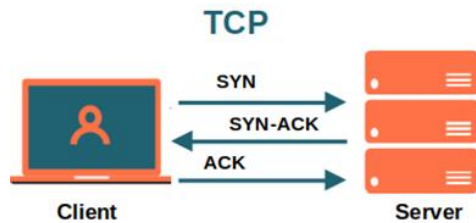


TCP/IP Protocol Suite

- Transport Layer
 - Transmission Control Protocol (TCP)
 - User Datagram Protocol (UDP)
- Network Layer
 - Internet Protocol (IP)
 - IPv4 (32 bits)
 - IPv6 (128 bits)

TCP vs UDP

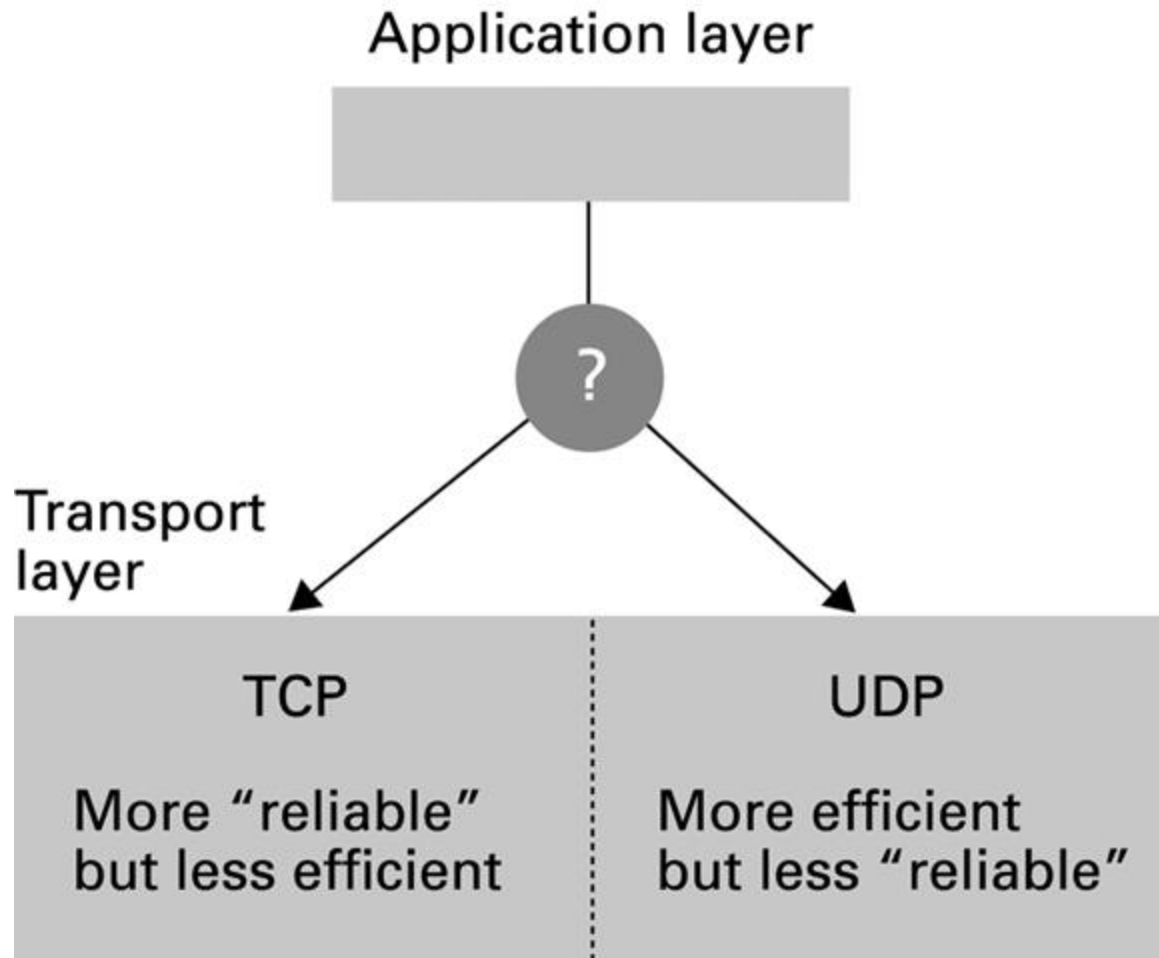
TCP vs UDP Communication



When to Use TCP and UDP



Figure 4.15 Choosing between TCP and UDP



Network protocols visually explained



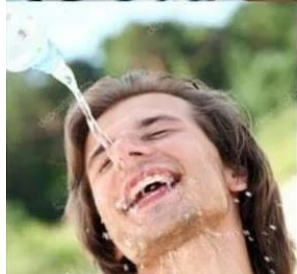
VPN



SSH



TCP



UDP



P2P



MULTICAST



BROADCAST

Security

- Attacks
 - Malware (viruses, worms, Trojan horses, spyware, phishing software)
 - Denial of service (DoS)
 - Spam
- Protection
 - VPN (Virtual Private Network)
 - Firewalls
 - Spam filters
 - Proxy Servers
 - Antivirus software

Encryption

- HTTPS and SSL
- Public-key Encryption
 - Public key: Used to encrypt messages
 - Private key: Used to decrypt messages
- Certificates and Digital Signatures
 - Certificate authorities

Figure 4.16 Public-key encryption

